

Prototype Project Reflection

For this section of the FYRE module, we were tasked with creating a prototype involving motors, homemade sensors, and other provided materials that provides a service. The most important part of our project was the planning. We started by drafting ideas that fit into the constraints and requirements of the project. We had a few different ideas, including a self-watering planter idea, but ultimately settled on a canopy system as it worked the best with the materials we were given. We created an in-depth plan on what we wanted our prototype to do, including things like inputs and outputs. We created flow charts and diagrams to illustrate how our project would work. From here, we created sketches of the physical design and how all the components would fit together. After we got the planning out of the way, we began the process of building our prototype.

While each person in the group had a focus on what they were working on, we all helped in other areas to create the finished product. We started with the base of the whole system, which we made out of popsicle sticks and represented our building. After our base was assembled, we moved on to the wiring, which we did ourselves by referencing previous notes. When our circuit was successfully built, we tested each component separately, like the sensors and LED. After we established everything was working, we prompted AI to write the code for how we wanted each piece to function, then had all the code added together to complete the working program. After our circuit was functioning with the code, we had to finish assembling our system. We created a longer arm to attach to the servo arm that would rotate about 90 degrees when activated and, when deactivated, retract and lay flush against the base. We used paper to create the canopy as it would be easy for it to fold in on itself. Now we had a fully functional retracting awning in place. We continued assembly by placing the breadboard inside the base and attaching cardboard to enclose it with an area for the LED to shine from.

We tested our fully assembled prototype. We hit the push button to manually activate the system to cover the human interaction requirement and also have a manual override to the system. When humid breath was directed toward the homemade humidity sensor, the LED turned on, indicating potential weather like rain, warning tenants to grab an umbrella or rain jacket before leaving. When water was detected on the moisture sensor, the servo arm rotated and the canopy deployed. When it was dry, the canopy retracted. Our trial was successful.

We met all the given requirements. We had multiple inputs, with our button and two sensors, and multiple outputs, with our LED and servo arm rotation. It provides a service to people and could potentially be marketed toward assisted living homes for people who need a little more support. Our prototype involved human interaction with the button, which manually activated the system.